

OSSP (Operating Systems And Systems Programming) - 25CS2104E

K.Rishita(2520030367)

1.)Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    char command[100];
```

```
    pid_t pid;
```

```
    printf("Enter a Linux command: ");
```

```
    scanf("%s", command);
```

```
    pid = fork();
```

```
    if (pid < 0)
```

```
{
```

```

    perror("fork failed");
    exit(1);
}

if (pid == 0)
{
    printf("\nChild Process\n");
    printf("Child PID: %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    execlp(command, command, (char *)NULL);

    perror("exec failed");
    exit(1);
}
else
{
    printf("\nParent Process\n");
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}

```

OUTPUT:

```
rishita@LAPTOP-Q85I28SL:~$ nano week1.c
rishita@LAPTOP-Q85I28SL:~$ gcc week1.c
rishita@LAPTOP-Q85I28SL:~$ ./a.out
Enter a Linux command: ls

Parent Process
Parent PID: 518
Child PID: 528

Child Process
Child PID: 528
Parent PID: 518
a.out      division.c  file2      gamelaumcher  notest.txt  subtraction.c
addition   excess.c    file5      gamelauncher2  p1,c.save  week1.c
addition.c excess1.c   filelauncher launcher      p2.c       week3.c
arithmetic excess2.c   filesys.c  multiplication p3.c       week4.c
arithmetic.c fill      forkdemo.c multiplication.c process     weekz4.c
arithmetic.sh file.txt   freefire  notes.txt     process.c
division   file1     game2     notes.txt2    subtraction
Child process completed.
rishita@LAPTOP-Q85I28SL:~$ |
```